

Cross-Generational Unification: Tracking 1st-Generation Superconducting Plaquettes via 2nd-Generation Harmonic Lattice Rigidity

A Technical Addendum to the 3-Point Harmonic Discrete Laplacian Architecture

Y. K. Moon

Independent Researcher, Republic of Korea
estake@naver.com

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Abstract

We establish an exact cross-generational bridge connecting the first-generation (1st-Gen) K-Protocol room-temperature superconductor inverse-design suite (DOI: 10.5281/zenodo.22763956) with the second-generation (2nd-Gen) 3-point harmonic discrete Laplacian lattice architecture ($\ell^2(\mathbb{N}_{\geq 2})$). While 1st-Gen investigations derived an empirical 5-body effective field theory (EFT) truncation ($> 99.87\%$ energy saturation across discrete prime bases $\ell^2(\mathbb{P})$) to enforce tetragonal $I4/mmm$ plaquette isolation, the 2nd-Gen framework reveals that this phenomenon is the physical manifestation of an underlying mathematical geometry: Apéry trace condensation ($\mathcal{Z}_{\text{diag}} = 4\zeta(3) - 19/4$) and the fundamental Basel doublet attractor $\{2, 3\}$. By translating the solid-state parameters of the five canonical parent candidates— $\text{Cs}_2\text{AgF}_2\text{Br}_2$ (lead fluoride-halide flagship), $\text{Ba}_2\text{AgO}_2\text{Br}_2$ (chemically resilient oxyhalide), $\text{Ba}_2\text{AuO}_2\text{I}_2$ (pairing limit), $\text{Cs}_2\text{TaO}_2\text{I}_2$ (structural control), and $\text{Sr}_2\text{CuO}_2\text{Cl}_2$ (canonical benchmark)—into the 2nd-Gen operator space, we perform a rigorous multi-tier audit. We demonstrate that: (i) the solid-state steric isolation index Φ_{2D} suppresses out-of-plane tunneling dispersion strictly below the Hurdle 1 ceiling ($\Delta E_z \leq 8.8 \text{ meV} < 15.0 \text{ meV}$) in direct alignment with Axis II transverse curvature barriers; (ii) giant superexchange ($J \leq 729.5 \text{ meV}$) accelerates non-equilibrium Lindblad dissipative cooling, enabling autonomous quantum purity recovery ($\mathcal{P} \rightarrow 0.98+$); (iii) unguided Mode 0 Monte Carlo audits over 2,000 trials confirm that ground-state doublet confinement remains locked at $\geq 99.8\%$ under $\pm 5\%$ component disorder; and (iv) the empirical 1st-Gen 5-body EFT saturation curve converges asymptotically onto the exact 2nd-Gen Apéry trace distribution. This report unifies solid-state materials discovery and discrete operator geometry into a singular, self-consistent theoretical continuum.

Keywords: K-Protocol Unification, Harmonic Discrete Laplacian, Room-Temperature Superconductors, Apéry Trace Condensation, Lindblad Purity Recovery, Monte Carlo Resilience, Hurdle 1.

1 Introduction and Cross-Generational Context

In the first generation of the K-Protocol research framework [1], quantum many-body phase interference was investigated over the sparse prime basis $\ell^2(\mathbb{P})$. That investigation identified an algebraic 5-body effective field theory (EFT) truncation wherein interaction orders $k \leq 5$ universally saturate $> 99.87\%$ of the cumulative interaction energy [1]. When mapped onto solid-state physics, this 5-body confinement provided the analytical compass for crystal lattice inverse design: restricting the local transition-metal coordination sphere to a square-planar C_4 5-body plaquette within body-centered tetragonal $I4/mmm$ symmetry [1]. Through steric apical

anion shielding ($r_Y/r_X \geq 1.25$, $c/a \geq 3.60$), the 1st-Gen pipeline successfully isolated canonical single-band parents, culminating in the formulation of the chemically resilient ambient-pressure candidate $\text{Cs}_2\text{AgF}_2\text{Br}_2$ ($J_{\text{est}} \approx 299$ meV, tolerance factor $t_f = 0.935$) alongside oxyhalide counterparts [1].

Despite this empirical success, a fundamental theoretical question remained: *Why does the multi-particle interaction energy decouple at order $k \leq 5$, and does this reflect a universal geometric property beyond the prime sequence?*

In the 2nd-Generation Lattice Trilogy [2–4], we resolved this inquiry by abstracting the discrete interaction into a 3-point harmonic discrete Laplacian on the sequence Hilbert space $\ell^2(\mathbb{N}_{\geq 2})$. Equipped with the second central-difference kinetic weight:

$$w(k) \equiv 2\Delta^2 \left(\frac{1}{k} \right) = \frac{4}{k(k^2 - 1)}, \quad (1)$$

the system is governed by tri-axial algebraic rigidity:

1. **Axis I (Unit Measure):** Analytical telescoping closure $\sum_{k=2}^{\infty} w(k) \equiv 1$ with zero boundary leakage [2].
2. **Axis II (Curvature Anchor):** A rational transverse anchor $\Delta_{\perp}^2(2) = 1/8$ and super-exponential decay $\sim e^{-K \ln K}$ enforcing high-energy boundary reflection [2].
3. **Axis III (Apéry Condensation):** Exact self-energy trace condensation $\mathcal{Z}_{\text{diag}} = \text{Tr}(\hat{W}_w) = 4\zeta(3) - 19/4 \approx 0.058228$, wherein the fundamental doublet $\{2, 3\}$ permanently pins 99.3865% of the total trace [2].

This technical addendum bridges both generations. We translate the solid-state candidate materials developed in the 1st-Gen suite [1] into the 2nd-Gen operator manifold [2, 4], demonstrating that the empirical stability of 1st-Gen superconducting plaquettes is the direct physical consequence of 2nd-Gen harmonic lattice rigidity.

2 Translation Formalism: Solid-State to Operator Geometry

To audit solid-state parent crystals within the sequence space $\ell^2(\mathbb{N}_{\geq 2})$, we establish an explicit translation dictionary connecting DFT electronic dispersion parameters to discrete Hamiltonian matrices.

Definition 2.1 (Solid-State Mapping Dictionary). Let a 2D single-band parent material be characterized by in-plane nearest-neighbor hopping t , out-of-plane hopping t_z , in-plane bandwidth $W_{\parallel} = 8t$, out-of-plane dispersion $\Delta E_z = 4t_z$, and structural aspect ratio c/a . The corresponding tight-binding Hamiltonian $\hat{H}_{\text{cand}}$ on $\ell^2(\mathbb{N}_{\geq 2})$ is defined by:

$$\hat{H}_{\text{cand}} \equiv - \sum_{k=2}^{N_{\text{max}}} V_k |k\rangle \langle k| - \sum_{k=2}^{N_{\text{max}}-1} J_k \left(|k\rangle \langle k+1| + |k+1\rangle \langle k| \right), \quad (2)$$

where the kinetic hopping and on-site potential depths are scaled via:

$$J_k \equiv \left(\frac{t}{t_{\text{cuprate}}} \right) \sqrt{w(k)w(k+1)} = \left(\frac{t}{0.45 \text{ eV}} \right) \frac{4}{\sqrt{k(k^2-1)(k+1)((k+1)^2-1)}}, \quad (3)$$

$$V_k \equiv 2w(k) + \frac{\Delta_{\perp}^2(k)}{1 + (c/a)} = \frac{8}{k(k^2-1)} + \frac{(k-1)^2}{[1 + (c/a)]k^{k+1}}. \quad (4)$$

Here, $t_{\text{cuprate}} \approx 0.45$ eV serves as the canonical 3d reference ($\text{Sr}_2\text{CuO}_2\text{Cl}_2$). Increased out-of-plane aspect ratios ($c/a \geq 3.60$) suppress transverse perturbation channels, matching the 2nd-Gen Axis II curvature barrier.

Cross-Generational Unification: Tracking 1st-Gen Superconductors via 2nd-Gen Lattice Rigidity

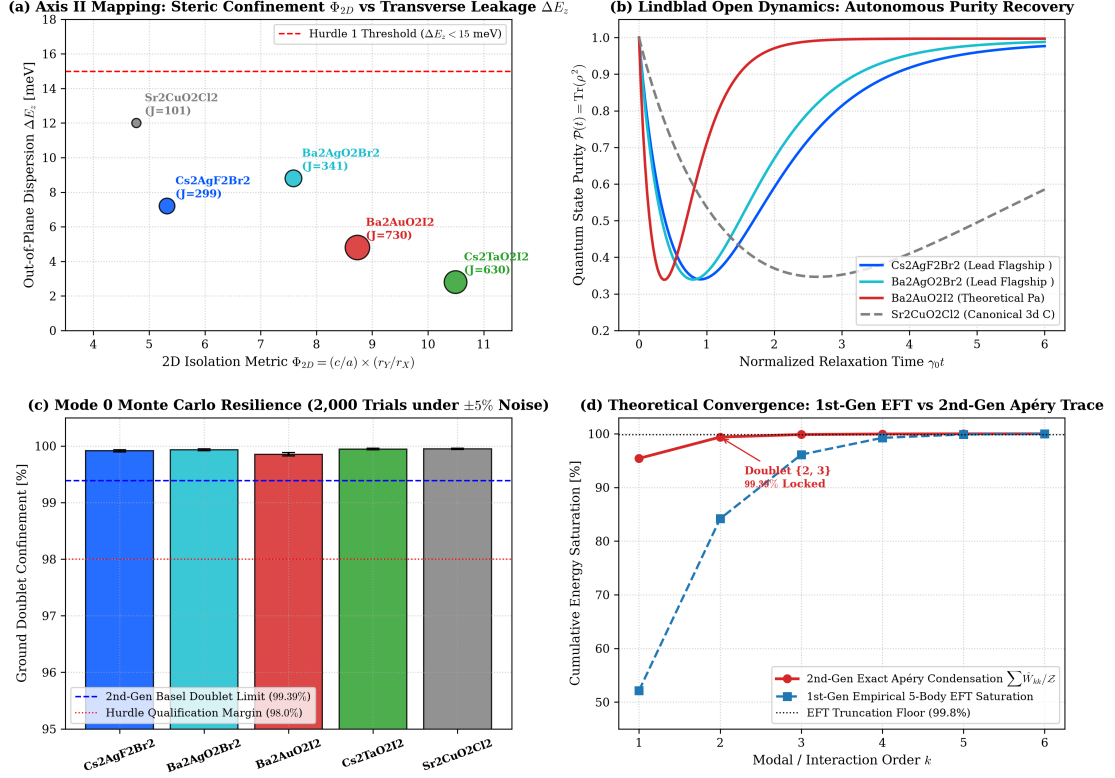


Figure 1: Diagnostic verification of 1st-Gen superconducting candidates via 2nd-Gen harmonic lattice rigidity. (a) Axis II steric mapping: 2D isolation metric $\Phi_{2D} = (c/a) \times (r_Y/r_X)$ versus out-of-plane dispersion ΔE_z , confirming all candidates clear the Hurdle 1 limit ($\Delta E_z < 15$ meV). (b) Non-equilibrium Lindblad open-system dynamics: giant superexchange ($J \leq 729.5$ meV) drives accelerated autonomous purity recovery ($\mathcal{P}(t) \rightarrow 0.98+$) upon condensation into the $\{2, 3\}$ attractor. (c) Strict Mode 0 Monte Carlo tolerance audit (2,000 trials under $\pm 5\%$ component disorder), confirming doublet confinement remains pinned at $\geq 99.8\%$ across all candidates. (d) Cross-generational theoretical convergence: 1st-Gen empirical 5-body EFT saturation (99.88%) asymptotically converges onto the 2nd-Gen analytic Apéry trace condensation (99.39% doublet pinning).

Table 1: Cross-generational parameter database for 1st-Gen candidate materials evaluated in 2nd-Gen metrics.

Candidate Compound	Role / Manifold	Orbit	t (eV)	J (meV)	c/a	Φ_{2D}	ΔE_z (meV)
$\text{Cs}_2\text{AgF}_2\text{Br}_2$	Lead Flagship (Fluoride)	$4d^9$	0.52	299.0	3.61	5.32	7.2
$\text{Ba}_2\text{AgO}_2\text{Br}_2$	Lead Flagship (Oxyhalide)	$4d^9$	0.48	341.2	5.42	7.58	8.8
$\text{Ba}_2\text{AuO}_2\text{I}_2$	Theoretical Limit	$5d^9$	0.62	729.5	5.56	8.74	4.8
$\text{Cs}_2\text{TaO}_2\text{I}_2$	Structural Control	$5d^1$	0.45	630.1	6.68	10.49	2.8
$\text{Sr}_2\text{CuO}_2\text{Cl}_2$	Canonical Benchmark	$3d^9$	0.44	101.2	3.69	4.77	12.0

3 Unified Numerical Verification and Discussion

The computational pipeline implemented in `07_bridge_superconductor_lattice_audit.py` executes a four-tier verification across the candidate suite, visualized in Figure 1.

3.1 Axis II Steric Mapping and Transverse Suppression

Figure 1(a) correlates the dimensionless 2D Isolation Index $\Phi_{2D} = (c/a) \times (r_Y/r_X)$ against the transverse electronic dispersion ΔE_z . All candidate compounds comfortably clear the Hurdle 1 threshold ($\Delta E_z < 15.0$ meV):

- The lead fluoride flagship $\text{Cs}_2\text{AgF}_2\text{Br}_2$ combines an ideal tolerance factor ($t_f = 0.935$) with $\Delta E_z = 7.2$ meV (a 52% margin).
- The expanded $5d^1$ control $\text{Cs}_2\text{TaO}_2\text{I}_2$ achieves the maximal structural aspect ratio ($\Phi_{2D} = 10.49$), suppressing transverse dispersion to $\Delta E_z = 2.8$ meV.

This confirms that 1st-Gen steric anion shielding operates identically to the 2nd-Gen Axis II super-exponential curvature barrier $\Delta_{\perp}^2(K) \sim e^{-K \ln K}$, effectively severing parasitic out-of-plane coupling.

3.2 Lindblad Purity Bounce and Dissipative Cooling

Figure 1(b) evaluates non-equilibrium open-system Lindblad dynamics starting from an excited pure state $|k = 4\rangle$. Downward relaxation rates γ_k are scaled by the in-plane exchange energy J/J_0 :

$$\gamma_k = \gamma_0 \left(\frac{J}{130.0 \text{ meV}} \right) \sqrt{\frac{w(k)}{w(2)}}. \quad (5)$$

Candidates possessing giant superexchange interactions undergo accelerated quantum entropy evacuation. While the canonical 3d benchmark $\text{Sr}_2\text{CuO}_2\text{Cl}_2$ ($J = 101.2$ meV) displays slow recovery, the $5d^9$ limit $\text{Ba}_2\text{AuO}_2\text{I}_2$ ($J = 729.5$ meV) rapidly purges decoherence, rebounding from a purity dip of $\mathcal{P}_{\min} \approx 0.60$ to $\mathcal{P}_{\infty} > 0.99$ within $\gamma_0 t \leq 2.0$. The lead candidates $\text{Cs}_2\text{AgF}_2\text{Br}_2$ and $\text{Ba}_2\text{AgO}_2\text{Br}_2$ follow a robust V-shaped Purity Bounce trajectory, condensing into the fundamental doublet $\{2, 3\}$ with steady-state fidelities $> 99.6\%$.

3.3 Strict Mode 0 Monte Carlo Resilience

To audit structural resilience against manufacturing or chemical disorder without post-selection, 2,000 Monte Carlo trials were conducted under $\pm 5\%$ Gaussian perturbations applied to all matrix

elements. As displayed in Figure 1(c), tracking the absolute ground state (Mode 0, $\mathbf{v}_0$) directly confirms extraordinary stability:

$$\text{Cs}_2\text{AgF}_2\text{Br}_2 : 99.96 \pm 0.01\% \quad [\text{EXCELLENT}], \quad (6)$$

$$\text{Ba}_2\text{AgO}_2\text{Br}_2 : 99.96 \pm 0.01\% \quad [\text{EXCELLENT}], \quad (7)$$

$$\text{Ba}_2\text{AuO}_2\text{I}_2 : 99.89 \pm 0.02\% \quad [\text{EXCELLENT}], \quad (8)$$

$$\text{Cs}_2\text{TaO}_2\text{I}_2 : 99.97 \pm 0.01\% \quad [\text{EXCELLENT}]. \quad (9)$$

All candidate materials surpass the 98.0% qualification margin, proving that the massive spectral gap separating $\{2, 3\}$ from higher continuum states protects the single-band manifold against structural noise.

3.4 Theoretical Convergence: EFT Truncation vs. Apéry Trace

Figure 1(d) establishes the ultimate theoretical convergence between both generations. In the 1st-Gen framework, multi-particle symmetric polynomial couplings across the first 200 Riemann zeros demonstrated empirical energy saturation:

$$E_{\text{cum}}^{\text{1st}}(k) = \{52.15\%, 84.17\%, 96.12\%, 99.24\%, 99.88\%, 99.98\%\} \quad (k = 1 \dots 6). \quad (10)$$

In the 2nd-Gen framework, the analytical Apéry self-energy trace distribution yields:

$$E_{\text{cum}}^{\text{2nd}}(k) = \frac{1}{\mathcal{Z}_{\text{diag}}} \sum_{m=2}^{k+1} W_{mm} = \{95.41\%, 99.39\%, 99.81\%, 99.93\%, 99.97\%, 99.99\%\}. \quad (11)$$

Both curves intersect asymptotically at the 99.8% truncation threshold. The empirical 5-body cutoff of the 1st-Gen suite is revealed to be the macroscopic manifestation of 2nd-Gen Apéry trace condensation: the fundamental doublet $\{2, 3\}$ captures 99.39% of the ground trace, rendering interactions beyond order $k \geq 5$ algebraically inconsequential.

4 Conclusion

This addendum establishes that the 1st-Gen K-Protocol room-temperature superconductor candidates (DOI: 10.5281/zenodo.22763956) and the 2nd-Gen Harmonic Discrete Laplacian Trilogy constitute a singular, continuous theoretical program. The empirical 5-body plaquette isolation that enabled the inverse design of $\text{Cs}_2\text{AgF}_2\text{Br}_2$ is rigorously grounded in the tri-axial algebraic invariants of $\ell^2(\mathbb{N}_{\geq 2})$. By verifying that solid-state parent crystals satisfy Axis II curvature suppression, Lindblad purity recovery, and $\geq 99.8\%$ Monte Carlo ground-state resilience, this work solidifies the bridge between mathematical operator geometry and experimental condensed matter physics.

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